TECHNOLOGY MANDATORY · FOOD AND AGRICULTURAL PRACTICES

# THE RIV BURGER

*Compact student workbook*

**READ · CHECK · RESET · EXPLAIN · APPLY**

| | |
|---|---|
| **Student** | |
| **Class** | |
| **Teacher** | |
| **Year** | |

Compact edition 2.0 · 50-page A4 edition · August 2026

COMPACT EDITION

## How to use this 50-page workbook

This edition preserves the complete ten-module learning journey while removing repeated instructions, oversized writing areas and separate single-purpose pages. Use the extended 113-page edition when students need more writing space or a slower paper pathway.

| | |
|---|---|
| **READ** | Three concise theory sections plus the module photograph and infographic. |
| **CHECK** | Six representative knowledge questions, then use the answer key to revisit theory. |
| **RESET** | Two course-specific Busy Work activities to vary the learning rhythm. |
| **EXPLAIN** | One combined response that connects the three theory sections. |
| **APPLY** | One focused project-evidence page linked to the Riv Burger design process. |

COURSE MAP

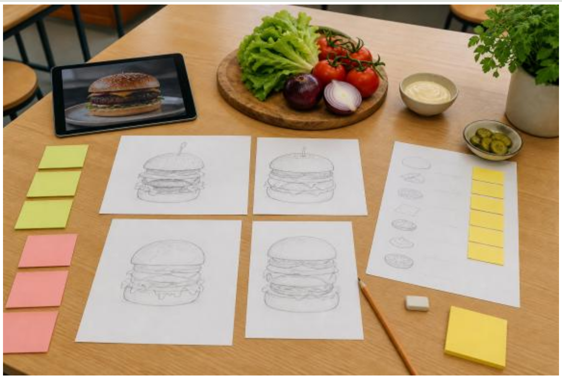
| Module | Focus | Compact evidence |
|---|---|---|
| 1 | **The challenge** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 2 | **Safe food practices** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 3 | **Tools and workflow** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 4 | **Burger anatomy** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 5 | **Ingredients and agriculture** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 6 | **Develop your design** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 7 | **Garden to kitchen** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 8 | **Plan production** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 9 | **Produce and present** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |
| 10 | **Evaluate and reflect** | 3 theory sections · 6 checks · 2 Busy Work activities · response · project evidence |

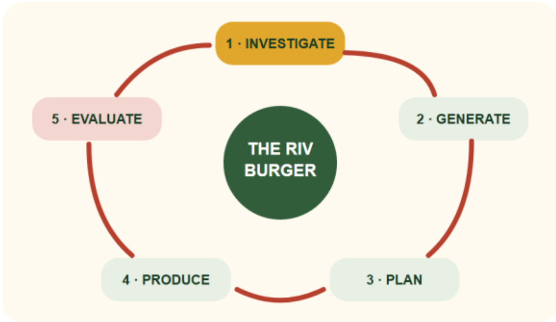
**Safety and submission boundary:** Follow current teacher directions for allergens, equipment, practical work, ingredients and supervision. Formal submission arrangements remain Teacher to confirm.

MODULE 1 · READ

# 1. The challenge

*Understand the design situation before proposing a solution.*






## 1.1 Design situation and brief

A design situation explains the need or opportunity. In this unit, a small local food business wants a burger that appeals to young people while showing that fresh, local and more sustainable choices can compete with familiar fast-food products. The design brief turns that situation into action: develop a signature burger featuring locally grown produce and home-grown herbs . Your solution needs to be realistic for a school food-technology setting.

## 1.2 Users and needs

A successful burger is designed for people, not only for a photograph. A young customer may value flavour, appearance, value, convenience and a size that is comfortable to hold and eat. The cook needs a clear recipe, safe method, suitable tools and enough time. These needs can compete. Extra height may look impressive but reduce ergonomics. Many fillings may add flavour but make production slower and the burger unstable.

## 1.3 Criteria and constraints

Criteria are the qualities used to judge success: function, appearance, creativity, ergonomics, ethical choices, environmental impact and time management. Constraints are real limits such as lesson time, available equipment, ingredient access, budget, skills, safety and dietary needs. Design habit: write criteria so they can be tested. “Easy to hold without fillings falling out” is more useful than “good”.

**Key terms:** design situation · brief · user · need · criterion · constraint · evidence

MODULE 1 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines design brief?**

| | | |
|---|---|---|
| A a clear design task developed from a need or opportunity | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A choose ingredients before understanding the need | B wait until evaluation before considering it | C identify the product, intended user and important qualities before developing ideas |

**3. Which statement best defines user need?**

| | | |
|---|---|---|
| A something the customer or producer requires from the solution | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A design only for a dramatic photograph | B wait until evaluation before considering it | C balance flavour, appearance, value, convenience, size and production needs |

**5. Which statement best defines testable criterion?**

| | | |
|---|---|---|
| A a specific quality that can be checked to judge design success | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A use “good” as the only success measure | B wait until evaluation before considering it | C write measurable or observable criteria and list realistic constraints |

### COMBINED GUIDED RESPONSE

1. Rewrite the design brief in your own words. Include the product, intended user and two important qualities. 2. Explain the needs of the intended customer and the cook. Identify one likely trade-off between those needs. 3. Write three testable success criteria and two realistic constraints for your Riv Burger.

> **Useful starters:** The task is to design a signature burger for… It needs to… · The intended customer needs… The cook needs… A trade-off occurs when… · My burger will be successful if… A constraint I must work within is…

Self-check: □ all three ideas addressed □ subject vocabulary used □ claims linked to reasons or evidence

MODULE 1 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 1 · Design language find-a-word

Find the words that help a designer turn a need into a clear, testable task.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| F | U | N | C | T | I | O | N | K | B | Q |
| R | C | W | N | N | T | N | K | E | F | Y |
| O | K | E | V | I | D | E | N | C | E | N |
| D | N | S | O | A | Z | O | A | S | I | D |
| R | W | G | Q | R | V | M | M | B | R | M |
| C | C | R | I | T | E | R | I | A | B | K |
| H | S | U | A | S | V | S | N | I | R | N |
| Q | K | K | R | N | E | H | U | N | U | M |
| E | U | K | X | O | O | D | C | E | A | U |
| L | P | R | C | C | N | T | B | N | H | K |
| T | G | L | M | N | Q | S | F | E | Q | F |

**BRIEF · USER · NEED · CRITERIA · CONSTRAINT · FUNCTION · EVIDENCE · DESIGN**

**Quick finish: Choose two found words and explain how they connect:** ______________________________ *Revisit: 1.1 Design situation and brief*

## BUSY WORK 2 · Brief sorter

Write N (need), U (user), CR (criterion) or CO (constraint) beside each statement.

| Statement or action | Answer |
|---|---|
| The burger should appeal to young people. | |
| The burger must be realistic in a school food-technology setting. | |
| The final burger should remain stable when held. | |
| A local business wants a fresh signature burger. | |
| Locally grown produce and home-grown herbs should feature. | |
| Lesson time and available equipment limit production. | |

**Quick finish: Which statement would be easiest to test, and why?** ________________________________ *Revisit: 1.2 Criteria and constraints*

MODULE 1 · APPLY

# Project evidence · The challenge

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### NEED, OPPORTUNITY AND INTENDED USER

#### THREE TESTABLE SUCCESS CRITERIA

#### IMPORTANT CONSTRAINTS

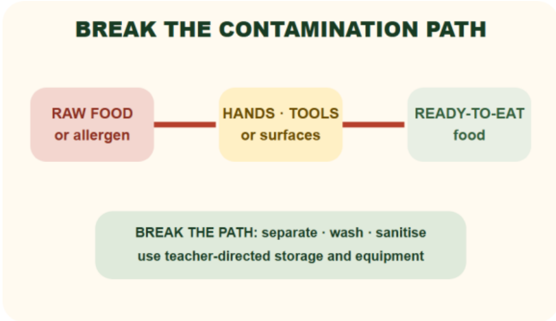
#### ONE EARLY BURGER DIRECTION AND WHY IT FITS

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 2 · READ

# 2. Safe food practices

*Protect people by controlling hygiene and food-production risks.*




## 2.1 Personal hygiene

Food handlers can transfer microorganisms, hair, dirt and allergens to food. Before production, secure hair, remove or control jewellery as directed, cover cuts, wear the required protective clothing, and wash hands thoroughly with soap and running water before drying them. Wash hands again after touching raw food, waste, your face or hair, and after any task that may contaminate them.

## 2.2 Cross-contamination

Cross-contamination happens when harmful microorganisms or allergens move from one food, surface, tool or person to another. Keep raw ingredients separate from ready-to-eat foods. Use the colour-coded boards and equipment directed by your teacher, clean and sanitise work areas, and never reuse a utensil that has touched raw food without washing it correctly. Non-negotiable: follow teacher instructions for storage, cooking, cooling and allergen controls. If unsure, stop and ask.

## 2.3 Hazards and controls

A hazard is something with the potential to cause harm. A risk control reduces the chance or severity of that harm. Knife controls include a stable board, correct grip, a safe cutting technique, full attention and carrying the knife only as instructed. Heat controls include dry oven mitts, clear handles and warning others about hot equipment. Safety also includes garden work: use gloves and follow instructions when handling potting mix, minimise dust, work in ventilation and wash hands afterwards.

**Key terms:** hygiene · allergen · contamination · hazard · risk · control · separate

MODULE 2 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines personal hygiene?**

| | | |
|---|---|---|
| A actions that prevent a food handler transferring contamination to food | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A wash hands only at the end of the lesson | B wait until evaluation before considering it | C secure hair, cover cuts, wear required protection and wash and dry hands correctly |

**3. Which statement best defines cross-contamination?**

| | | |
|---|---|---|
| A the transfer of harmful microorganisms or allergens between food, people, tools or surfaces | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A reuse a raw-food utensil for salad without washing it | B wait until evaluation before considering it | C separate raw and ready-to-eat foods and use cleaned, sanitised equipment |

**5. Which statement best defines risk control?**

| | | |
|---|---|---|
| A an action that reduces the chance or severity of harm | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A name a hazard without changing how the task is performed | B wait until evaluation before considering it | C match each identified hazard with a specific control |

## COMBINED GUIDED RESPONSE

1. Write your before-cooking hygiene routine as a clear six-step checklist. 2. Explain how cross-contamination could occur during burger production and give three controls that break the pathway. 3. Identify one biological, one physical and one heat or equipment hazard. Give a specific control for each.

> **Useful starters:** Before cooking I will: 1… 2… 3… · Cross-contamination could occur when… This pathway is controlled by… · One biological hazard is… I will control it by…

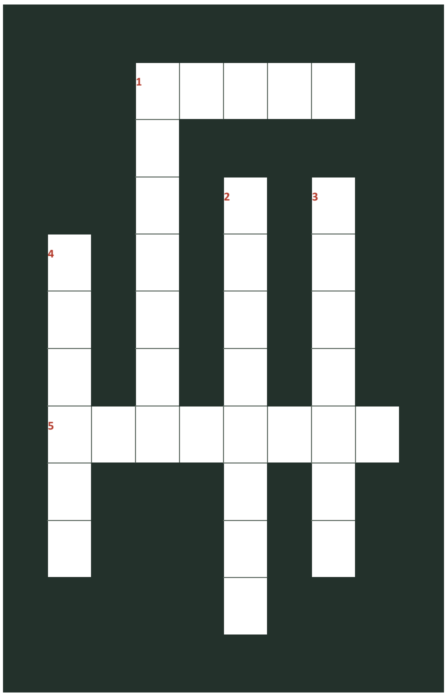
Self-check: ◻ all three ideas addressed ◻ subject vocabulary used ◻ claims linked to reasons or evidence

MODULE 2 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 3 · Food-safety crossword

Use the clues to complete the food-safety crossword.




Across: 1 Free from visible dirt and unwanted contamination. | 5 A substance that can trigger an allergic reaction.
Down: 1 An action used to reduce a risk. | 2 Keep raw and ready-to-eat food apart. | 3 Personal and workplace practices that help keep food safe. | 4 Something with the potential to cause harm.

**Quick finish: Circle the answer most directly linked to preventing cross-contamination.** *Revisit: 2.1 Hygiene, allergens and contamination*

## BUSY WORK 4 · Hazard-control match

Match each situation A-F with the most specific control 1-6.

| A-F | 1-6 |
|---|---|
| A. Raw patty mixture and salad share one board. | 1. Stop, identify the knife and wash it separately. |
| B. A spill is left in a walkway. | 2. Tie hair back before beginning. |
| C. A wet oven mitt is beside a hot tray. | 3. Use separate cleaned equipment for ready-to-eat food. |
| D. An ingredient label is not checked. | 4. Check the label and follow the teacher's allergen process. |
| E. A knife is hidden under washing-up water. | 5. Clean and dry the spill promptly and warn others. |
| F. Long hair is loose near food. | 6. Replace it with a dry mitt and follow the hot-equipment routine. |

**Matches: A-__ B-__ C-__ D-__ E-__ F-__**

**Quick finish: Which control requires you to stop and ask the teacher? ________________________________** *Revisit: 2.2 Managing hazards and risk*

MODULE 2 · APPLY

# Project evidence · Safe food practices

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### PERSONAL HYGIENE ROUTINE BEFORE PRODUCTION

#### RAW FOOD TO READY-TO-EAT CONTAMINATION PATH AND CONTROLS

#### ONE ALLERGEN CHECK AND ONE PHYSICAL OR HEAT HAZARD CONTROL

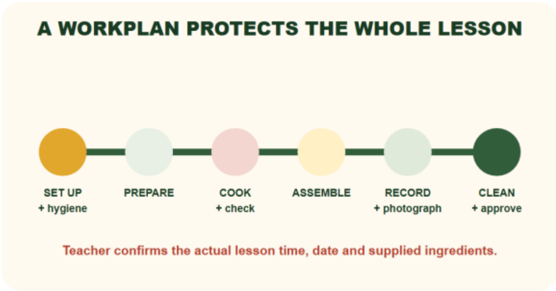
#### WHEN I MUST STOP AND ASK THE TEACHER

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 3 · READ

# 3. Tools and workflow

*Prepare an efficient workstation and use tools for their intended purpose.*




### 3.1 Mise en place

Mise en place means putting everything in place before production. Read the complete method, wash hands, sanitise the bench, collect and check equipment, measure ingredients, prepare waste containers and preheat equipment only when instructed. This preparation reduces rushing and helps you notice missing ingredients or unsafe conditions before they become problems.

### 3.2 Selecting tools

Select a tool because its function suits the job. A cook's knife can chop larger ingredients; a small knife provides control for smaller preparation; measuring spoons suit small quantities; a digital scale measures mass; and a spatula can turn a patty while keeping hands away from heat. Use every tool only after demonstration and according to classroom rules.

### 3.3 A clean workflow

Plan movement from preparation to cooking, assembly and cleaning. Keep handles away from bench edges, wipe spills promptly, return ingredients to safe storage, and clean as you go without placing sharp tools into hidden washing-up water. At the end, wash, rinse and dry equipment as directed, sanitise the workstation and complete the teacher's practical checklist.

> **Key terms:** mise en place · workflow · function · technique · clean-as-you-go

MODULE 3 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines mise en place?**

| | | |
|---|---|---|
| A putting ingredients, equipment and the work area in place before production | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A start cooking before checking the complete method | B wait until evaluation before considering it | C read the method, prepare the bench, collect tools and measure ingredients |

**3. Which statement best defines tool selection?**

| | | |
|---|---|---|
| A choosing equipment because its function suits the required task | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A choose a tool only because it is nearby | B wait until evaluation before considering it | C match the cook's knife, small knife, scale, spoons or spatula to its intended job |

**5. Which statement best defines clean workflow?**

| | | |
|---|---|---|
| A an ordered movement from preparation through cooking, assembly and cleaning | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A hide sharp tools in washing-up water | B wait until evaluation before considering it | C clean as you go, manage spills and return ingredients to safe storage |

COMBINED GUIDED RESPONSE

1. Describe how you would set up a safe workstation for preparing a burger patty and fresh salad. 2. Choose four tools for your likely recipe. Explain the function of each and one safe-use point. 3. Plan a clean workflow from preparation to final workstation check. Explain how it prevents two problems.

**Useful starters:** I will begin by… The raw-food area will… The salad area will… · I will use a… to… A safe-use point is… · My workflow moves from… to… Cleaning… prevents…

Self-check: ☐ all three ideas addressed ☐ subject vocabulary used ☐ claims linked to reasons or evidence

MODULE 3 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 5 · Mise en place shuffle

Number the actions 1-8 to create a calm preparation sequence. Teacher directions always come first.

| Statement or action | Answer |
|---|---|
| Gather the teacher-approved ingredients and equipment. | |
| Read the recipe and current teacher instructions. | |
| Begin the first preparation task in the workplan. | |
| Wash hands and prepare personal hygiene requirements. | |
| Check that the work area is ready and hazards are controlled. | |
| Measure ingredients and organise them for use. | |
| Confirm the correct boards, knives and other tools. | |
| Keep the bench organised as production begins. | |

**Quick finish: Why does reading the plan come before collecting equipment? ________________________** *Revisit: 3.1 Mise en place and workflow*

## BUSY WORK 6 · Tool-to-job match

Match each tool A-F with its best job 1-6. Then underline the job that also needs a safety check.

| A-F | 1-6 |
|---|---|
| A. Digital scale | 1. Turn or lift food using the planned technique. |
| B. Spatula | 2. Measure a small ingredient quantity. |
| C. Chopping board | 3. Protect hands when handling teacher-approved hot equipment. |
| D. Measuring spoon | 4. Weigh ingredients accurately. |
| E. Oven mitt | 5. Provide a stable preparation surface while preventing contamination. |
| F. Chef's knife | 6. Cut ingredients using the demonstrated grip and technique. |

**Matches: A-__ B-__ C-__ D-__ E-__ F-__**

**Quick finish: One setup check I would make before using a knife is: ______________________________** *Revisit: 3.2 Tools, equipment and safe technique*

MODULE 3 · APPLY

# Project evidence · Tools and workflow

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### MISE EN PLACE: INGREDIENTS, EQUIPMENT AND WORKSTATION

#### FOUR TOOLS: INTENDED JOB AND SAFE-USE POINT

#### WORKFLOW FROM PREPARATION TO CLEANING

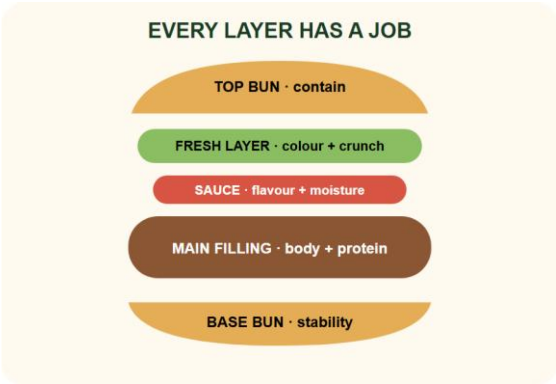
#### ONE PROBLEM THIS WORKFLOW PREVENTS

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 4 · READ

# 4. Burger anatomy

*Use structure and sensory properties to make the product work.*




## 4.1 A layered food system

A burger is a system of interacting parts. The bun contains the product and gives structure. The main filling provides body and a central flavour. Fresh components add colour, texture and freshness. Sauces add moisture and flavour but can also make the bun soggy. Each layer affects the whole product. Top bun · hold and protect Fresh layer · colour and crunch Sauce · flavour and moisture Main filling · body and protein Base bun · stable foundation

## 4.2 Sensory properties

Sensory properties are features detected by the senses: appearance, aroma, flavour, texture and sound. Designers can create contrast—such as a crisp vegetable with a soft bun—but too many competing flavours or textures can make the product confusing. Use precise words in evaluation: golden, crisp, juicy, smoky, fresh, balanced or overly salty are more useful than “good”.

## 4.3 Function and stability

Place wet ingredients away from surfaces that become soggy quickly, drain washed vegetables, match the bun size to the filling, and avoid layers that slide against each other. The base should sit flat and the assembled burger should remain comfortable to hold. A product can look appealing and still fail if it collapses, leaks or is difficult to eat.

**Key terms:** layer · function · sensory · texture · flavour · stability · ergonomics

MODULE 4 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines layered food system?**

| | | |
|---|---|---|
| A interacting burger parts in which each layer affects the whole product | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A add layers without considering how they interact | B wait until evaluation before considering it | C give every layer a clear function and arrange it deliberately |

**3. Which statement best defines sensory property?**

| | | |
|---|---|---|
| A a product feature detected through appearance, aroma, flavour, texture or sound | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A evaluate every feature only as “good” | B wait until evaluation before considering it | C use exact sensory vocabulary and plan complementary contrasts |

**5. Which statement best defines burger stability?**

| | | |
|---|---|---|
| A the ability of the assembled product to remain together and comfortable to hold | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A stack several wet, sliding layers directly on the base | B wait until evaluation before considering it | C drain vegetables, match component sizes and position wet or slippery layers carefully |

## COMBINED GUIDED RESPONSE

1. Plan a burger stack from bottom to top. Explain the function of each layer. 2. Write a sensory profile for your intended burger using precise appearance, aroma, flavour and texture words. 3. Explain three design decisions that will improve your burger’s stability and ease of eating.

> **Useful starters:** From the base upward, my burger contains... The base layer functions to... · The burger will appear... Its aroma will... The flavour will... The texture will... · I will improve stability by... This prevents...

Self-check: ☐ all three ideas addressed ☐ subject vocabulary used ☐ claims linked to reasons or evidence

MODULE 4 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 7 · Burger layer find-a-word

Find the terms used to explain how burger layers work together.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| R | O | J | M | N | R | B | Z | X | L | P |
| H | N | Q | K | F | I | H | Y | H | E | F |
| C | G | G | V | R | Y | A | T | B | I | L |
| N | R | U | A | W | G | A | I | L | P | A |
| U | E | F | P | M | V | D | L | C | G | V |
| O | T | X | Q | W | O | I | I | Y | H | O |
| U | R | E | Y | E | N | R | B | S | S | U |
| I | G | T | E | G | X | Q | A | B | E | R |
| I | O | X | V | T | E | X | T | U | R | E |
| T | W | O | E | C | U | A | S | N | F | E |
| K | K | O | C | K | D | L | L | S | H | C |

**BUN · FILLING · SAUCE · FRESH · STABILITY · TEXTURE · AROMA · FLAVOUR**

**Quick finish: Which two words are most closely linked to whether a burger is easy to hold? ___________** *Revisit: 4.1 Burger systems and layer functions*

## BUSY WORK 8 · Stack stability trouble-shooter

For each burger problem, select the best first design response and explain your choice.

| Statement or action | Answer |
|---|---|
| The filling slides out when the burger is lifted. | |
| The bun becomes soggy before serving. | |
| Every layer feels soft and similar. | |
| The burger is too tall for the intended user. | |

**Quick finish: Which response changes function without automatically changing flavour? _______________** *Revisit: 4.2 Sensory and functional analysis*

MODULE 4 · APPLY

# Project evidence · Burger anatomy

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### LAYER ORDER FROM BASE TO TOP

#### FUNCTION AND SENSORY EFFECT OF EACH MAIN LAYER

#### LIKELY STABILITY OR ERGONOMIC PROBLEM

#### SPECIFIC DESIGN CHANGE THAT ADDRESSES THE PROBLEM

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 5 · READ

# 5. Ingredients and agriculture

*Trace ingredients to industries, producers and places.*


## 5.1 From primary production to plate

Agriculture supplies plant and animal products. Wheat may be milled into flour for buns; cattle or legumes may supply a patty ingredient; dairy farming supplies milk for cheese; and horticulture supplies vegetables and herbs. Processing, packaging, transport, retail and food service connect primary production to the burger.

## 5.2 Local and seasonal choices

"Local" should be supported by evidence, not used as a vague label. It may refer to a grower, producer or business within a defined region. Seasonal produce can be plentiful and suitable for local conditions, but availability and price still need checking. Research the producer, ingredient origin, distance, season and the evidence used to make each claim.

## 5.3 Ethical and environmental factors

Food choices can affect soil, water, biodiversity, animal welfare, energy use, packaging, transport and waste. No ingredient is automatically perfect. A responsible comparison considers reliable evidence and the full production system. Practical actions include planning quantities, using suitable portions, choosing reusable equipment, separating waste as directed and using garden produce effectively.

> **Key terms:** primary production · process · transport · local · seasonal · evidence

MODULE 5 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines food system?**

| | | |
|---|---|---|
| A the connected stages from primary production through processing, transport, retail and service | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A treat the supermarket as the original source of every ingredient | B wait until evaluation before considering it | C link each ingredient to its agricultural industry and later journey stages |

**3. Which statement best defines local food claim?**

| | | |
|---|---|---|
| A a claim supported by a defined region and evidence of producer or ingredient origin | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A call an ingredient local because its packaging is green | B wait until evaluation before considering it | C define the region and verify producer, origin, distance and season |

**5. Which statement best defines responsible food choice?**

| | | |
|---|---|---|
| A a choice that considers environmental, ethical and practical effects across the production system | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A declare an ingredient perfect without investigating its production system | B wait until evaluation before considering it | C compare evidence about soil, water, welfare, energy, packaging, transport and waste |

### COMBINED GUIDED RESPONSE

1. Choose four likely burger ingredients. Trace each to an agricultural industry and identify one processing or transport step. 2. Define what “local” will mean in your design, then record the evidence you would need to support two local ingredient claims. 3. Compare two possible ingredients using two environmental or ethical factors, then justify the more responsible choice.

> **Useful starters:** The ingredient… comes from the… industry. Before use it is… · For this project, local means… I would verify… by… · I compared… and… using… Although…, I would choose… because…

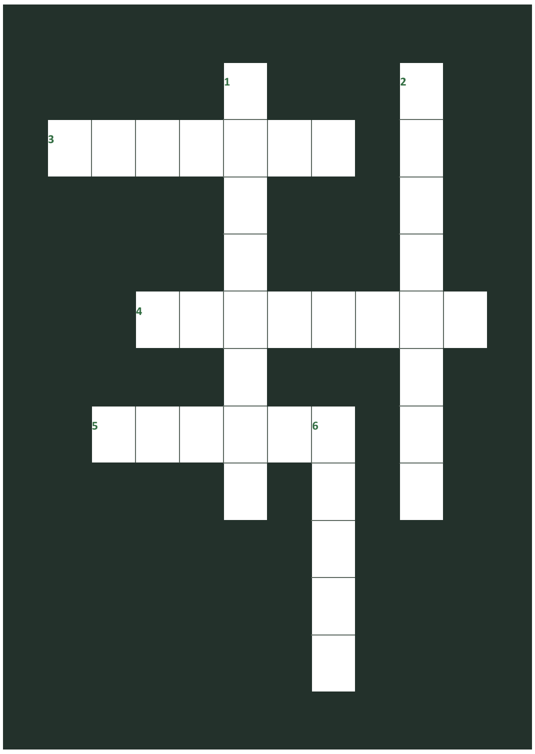
Self-check: ☐ all three ideas addressed ☐ subject vocabulary used ☐ claims linked to reasons or evidence

MODULE 5 · BUSY WORK

# Puzzle breaks: practise and revisit

### BUSY WORK 9 · Ingredient journey crossword

Complete the crossword to reveal the stages and evidence used in an ingredient journey.




Across: 3 A set of actions that changes or prepares a food. | 4 A person or business that grows or makes food. | 5 The stage where products are sold to customers.
Down: 1 Available at a particular time of year. | 2 Information used to support a claim. | 6 A word that needs a defined distance or region before it is a useful claim.

**Quick finish: Put a star beside the answer needed to support a local-food claim.** *Revisit: 5.1 From farm to plate*

### BUSY WORK 10 · Claim detective

Write S (supported) or V (vague). If vague, note the evidence that is missing.

| Statement or action | Answer |
|---|---|
| This herb came from our school garden; the harvest record is attached. | |
| This is definitely the most sustainable ingredient. | |
| The producer's current website identifies the NSW growing region. | |
| Local food is always better for the environment. | |
| The package names the producer and place of origin. | |
| It looks fresh, so it must have travelled the shortest distance. | |

**Quick finish: Rewrite one vague claim so it could be checked: ______________________________________** *Revisit: 5.2 Local, seasonal and sustainable choices*

MODULE 5 · APPLY

# Project evidence · Ingredients and agriculture

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### TWO LIKELY INGREDIENTS AND THEIR AGRICULTURAL INDUSTRIES

#### PRODUCTION JOURNEY: GROW OR RAISE, PROCESS, TRANSPORT AND RETAIL

#### DEFINE LOCAL FOR THIS INVESTIGATION

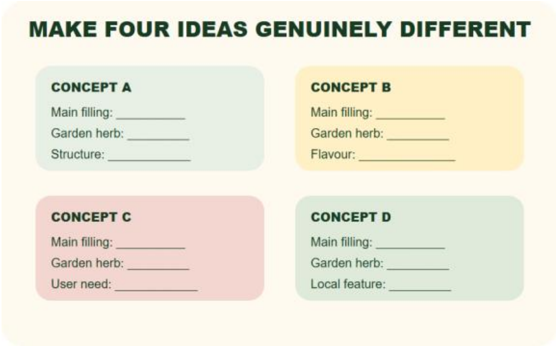
#### EVIDENCE SOURCE AND ONE RESPONSIBLE-CHOICE TRADE-OFF

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 6 · READ

# 6. Develop your design

*Generate alternatives, test them against criteria and justify a direction.*




## 6.1 Divergent thinking

Do not stop at the first idea. Generate several distinct burger concepts by changing the main filling, bun, herbs, vegetables, sauce, form or flavour profile. Four genuinely different options create a stronger comparison than four drawings of almost the same product.

## 6.2 Annotating concepts

An annotation explains a design decision. Label ingredients, layer order, sensory effect, source, function, safety issue or expected user benefit. “Parsley yoghurt sauce adds freshness and uses a garden herb” communicates more than an arrow labelled “sauce”.

## 6.3 Choosing with evidence

Use your criteria to compare options. A decision matrix can make trade-offs visible, but scores need reasons. The highest total does not remove your responsibility to explain why the chosen design best fits the brief and constraints. Appropriate response example: “I selected Concept C because its flat patty and matching bun improve stability, while garden parsley supports the brief. I will reduce the sauce quantity to control sogginess.”

> **Key terms:** divergent thinking · concept · annotation · criterion · decision · trade-off

MODULE 6 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines divergent thinking?**

| | | |
|---|---|---|
| A generating several meaningfully different responses before selecting one | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A redraw the same burger four times with different labels | B wait until evaluation before considering it | C change filling, bun, herbs, vegetables, sauce, form or flavour direction |

**3. Which statement best defines design annotation?**

| | | |
|---|---|---|
| A a note that explains the reason, function or expected effect of a design decision | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A draw an arrow and write only “sauce” | B wait until evaluation before considering it | C label ingredients, layer order, sensory effect, source, safety and user benefit |

**5. Which statement best defines evidence-based selection?**

| | | |
|---|---|---|
| A choosing a concept by comparing it with criteria and explaining trade-offs | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A select the highest total without checking the reasons behind it | B wait until evaluation before considering it | C score options, give reasons and identify a needed refinement |

COMBINED GUIDED RESPONSE

1. Describe four distinct burger concepts. For each, identify the main filling, local or garden feature and flavour direction. 2. Write six purposeful annotations for one concept. Cover function, sensory effect, sourcing and a user benefit. 3. Select one concept. Justify it against at least four criteria and identify one change needed before production.

> **Useful starters:** Concept A uses… Its local or garden feature is… The flavour direction is… · The… is positioned here because… This gives the user… · I selected Concept… because it best meets… It needs one change:…

Self-check: ☐ all three ideas addressed ☐ subject vocabulary used ☐ claims linked to reasons or evidence

MODULE 6 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 11 · Concept variation find-a-word

Find the features that can be changed deliberately when generating burger concepts.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| S | G | P | J | U | S | S | R | H | P | Z |
| F | L | E | L | B | A | T | E | G | E | V |
| K | L | R | H | U | C | R | U | N | V | U |
| U | H | A | C | A | B | U | T | I | T | H |
| Q | V | B | V | C | S | C | L | L | X | C |
| X | D | N | U | O | A | T | L | L | Q | J |
| T | U | C | D | N | U | U | H | I | Y | O |
| L | Q | N | U | C | C | R | E | F | W | O |
| S | C | B | Z | E | E | E | Y | F | U | B |
| I | I | A | B | P | F | H | P | W | T | K |
| B | F | E | J | T | T | N | A | Y | M | L |

FILLING · BUN · HERB · VEGETABLE · SAUCE · FLAVOUR · STRUCTURE · CONCEPT

**Quick finish: Name three features you would change to make two concepts genuinely different: ________** *Revisit: 6.1 Generating design options*

## BUSY WORK 12 · Concept quick-test

Read each design move. Decide whether it strengthens the concept (S), weakens it (W), or needs more evidence (E).

| Statement or action | Answer |
|---|---|
| The student checks the chosen concept against four written criteria. | |
| The final option is chosen only because it is the student's favourite. | |
| A local ingredient claim is made without naming a source or region. | |
| The student compares stability, sensory qualities and production limits. | |
| All four concepts change only the sauce colour. | |

**Quick finish: What evidence would turn the third move into a stronger decision? ______________________** *Revisit: 6.2 Selecting and justifying a concept*

MODULE 6 · APPLY

# Project evidence · Develop your design

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### CONCEPT A: FILLING, BUN, HERB OR LOCAL FEATURE, FLAVOUR AND STRUCTURE

#### CONCEPT B: MAKE IT GENUINELY DIFFERENT

#### COMPARE BOTH CONCEPTS AGAINST FOUR CRITERIA

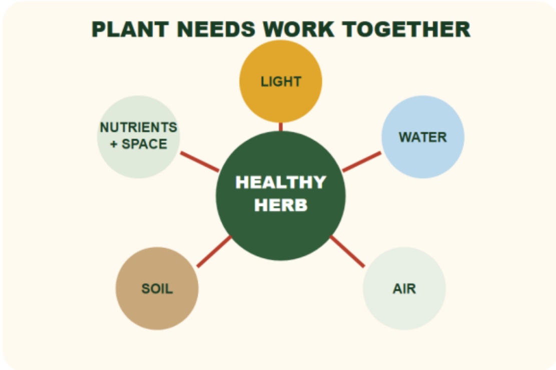
#### SELECTED CONCEPT, EVIDENCE AND ONE REFINEMENT

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 7 · READ

# 7. Garden to kitchen

*Connect plant needs, herb production and responsible resource use.*




## 7.1 What plants need

Healthy plants require light, water, air, suitable temperature, nutrients and a growing medium that supports roots. Too little water can cause wilting; too much can reduce oxygen around roots. Light and temperature affect growth, while nutrients support plant processes.

## 7.2 Growing and harvesting herbs

Herbs such as parsley, basil or chives can add flavour, aroma and colour. Identify plants correctly, harvest only with permission, select healthy material, wash produce as instructed and use clean tools. Harvesting methods should allow the plant to continue growing where possible.

## 7.3 Sustainability in practice

Sustainable practice aims to meet present needs while protecting future environmental, social and economic wellbeing. In a school garden and kitchen, this includes efficient watering, caring for soil, planning harvests, avoiding unnecessary food waste and valuing the work required to grow ingredients. A claim should connect an action to an effect—for example, measuring ingredients carefully can reduce avoidable waste.

**Key terms:** light · water · air · soil · nutrients · spacing · harvest · sustainability

MODULE 7 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines plant need?**

| | | |
|---|---|---|
| A a condition or resource required for healthy growth | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A add more water whenever a plant looks unhealthy | B wait until evaluation before considering it | C manage light, water, air, temperature, nutrients and root conditions together |

**3. Which statement best defines responsible harvesting?**

| | | |
|---|---|---|
| A collecting correctly identified healthy plant material while protecting safety and future growth | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A taste an unknown plant to identify it | B wait until evaluation before considering it | C gain permission, identify the herb, use clean tools and wash produce as directed |

**5. Which statement best defines sustainable practice?**

| | | |
|---|---|---|
| A meeting present needs while protecting future environmental, social and economic wellbeing | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A harvest more produce than the recipe can use | B wait until evaluation before considering it | C water efficiently, care for soil, plan harvests and measure ingredients |

### COMBINED GUIDED RESPONSE

1. Explain how light, water, air, nutrients and soil or growing medium support a healthy herb plant. 2. Describe the safe path for an approved herb from the school garden to your burger. 3. Propose three realistic garden-to-kitchen sustainability actions and explain the likely effect of each.

**Useful starters:** Light supports… Water is needed for… The growing medium… · First, the herb must be… After permission is given… Before use… · One action is… This is likely to… because…

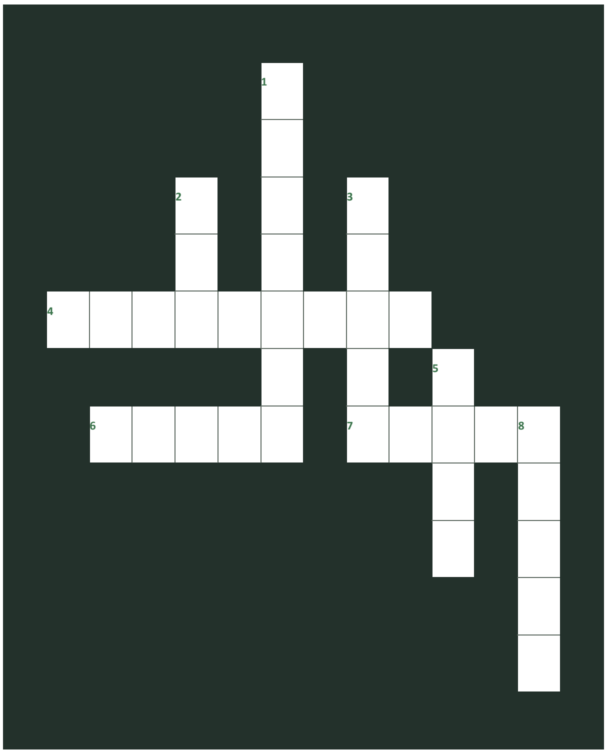
Self-check: □ all three ideas addressed □ subject vocabulary used □ claims linked to reasons or evidence

MODULE 7 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 13 · Plant-needs crossword

Complete the crossword using the factors that support plant growth and responsible harvesting.



Across: 4 Substances plants need for healthy growth. | 6 Energy source needed for plant growth. | 7 Plant structures that take up water and help anchor the plant.
Down: 1 Collecting usable plant material at an appropriate time. | 2 Needed around leaves and roots. | 3 Needed in suitable amounts; too much or too little can cause problems. | 5 Growing medium that can hold water, air and nutrients. | 8 Room that reduces competition between plants.

**Quick finish: Circle the two answers most directly affected when plants are overcrowded.** *Revisit: 7.1 Plant needs and herb growth*

## BUSY WORK 14 · Garden problem solver

Match each observation A-F with the most sensible first investigation 1-6. Ask the teacher before changing the garden.

| A-F | 1-6 |
|---|---|
| A. Several plants are crowded together. | 1. Check drainage and the watering routine. |
| B. Leaves are wilting. | 2. Review spacing and competition. |
| C. Growth is leaning towards one side. | 3. Check light direction and access. |
| D. Soil stays very wet. | 4. Review the harvesting amount and timing. |
| E. Leaves show unexpected damage. | 5. Check water availability and recent conditions. |
| F. Herbs are being removed faster than they regrow. | 6. Observe carefully and seek teacher guidance before acting. |

**Matches: A-__ B-__ C-__ D-__ E-__ F-__**

**Quick finish: Why is observation needed before changing the garden? ______________________________** *Revisit: 7.2 Garden planning and responsible harvesting*

MODULE 7 · APPLY

# Project evidence · Garden to kitchen

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### CHOSEN HERB: IDENTIFICATION, EDIBLE PART AND FLAVOUR

#### LIGHT, WATER, SOIL, AIR, NUTRIENTS AND SPACING NEEDS

#### PERMISSION, RESPONSIBLE HARVESTING AND KITCHEN PREPARATION

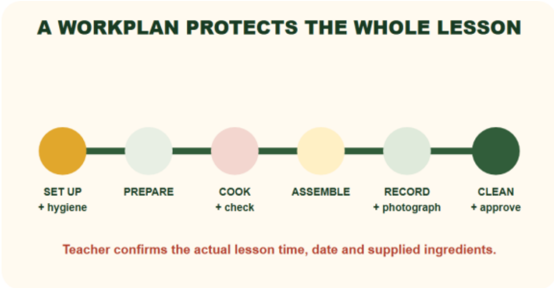
#### ONE GARDEN-TO-KITCHEN SUSTAINABILITY ACTION AND ITS EFFECT

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 8 · READ

# 8. Plan production

*Turn the selected concept into an accurate, manageable production plan.*




## 8.1 A production-ready recipe

A recipe must communicate exact ingredients, quantities and an ordered method. Use consistent units, name preparation requirements such as “finely chopped”, and include the yield. Sequence steps so preparation, cooking and assembly can happen safely within the available time.

## 8.2 Food order and resources

A food order totals the ingredients required for the planned yield. Check what the school supplies and what must be ordered or brought, according to teacher directions. Avoid duplicates, use realistic quantities and identify any ingredient with an allergen or storage requirement.

## 8.3 Workplan

A workplan maps time, activity, equipment and ingredients. Begin with hygiene and setup, identify tasks that can overlap safely, allow time for cooking and checking, and protect enough time for assembly, photography, cleaning and evaluation. Dates, lesson length, supplied ingredients and submission arrangements are Teacher to confirm .

> **Key terms:** recipe · yield · quantity · method · food order · resource · workplan

MODULE 8 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines production-ready recipe?**

| | | |
|---|---|---|
| A exact ingredients, quantities, yield and an ordered method that another cook can follow | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A list ingredients without quantities or method order | B wait until evaluation before considering it | C use consistent units, preparation words and safe sequence |

**3. Which statement best defines food order?**

| | | |
|---|---|---|
| A a totalled list of ingredients required for the planned yield | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A copy every possible ingredient without checking the final recipe | B wait until evaluation before considering it | C check supplies, total quantities and flag allergens or storage needs |

**5. Which statement best defines workplan?**

| | | |
|---|---|---|
| A a timed map of activities, equipment and ingredients across the practical | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A plan only the cooking and assembly steps | B wait until evaluation before considering it | C sequence tasks, identify safe overlaps and protect evidence and cleaning time |

### COMBINED GUIDED RESPONSE

1. Draft your recipe with yield, exact quantities and a numbered method. Include preparation words and safe sequencing. 2. Prepare a food order for your final recipe. Show totals, supply responsibility and any allergen or storage note. 3. Create a timed workplan from setup to cleaning. Identify equipment, ingredients and one quality check at each major stage.

> **Useful starters:** Yield:... Ingredients:... Method: 1... · Ingredient... Total required... Supplied by... Special note... · At... minutes I will... using... I will check...

Self-check: □ all three ideas addressed □ subject vocabulary used □ claims linked to reasons or evidence

MODULE 8 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 15 · Recipe language find-a-word

Find the words that make a recipe and production plan precise enough to follow.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| I | D | T | I | A | K | S | L | S | P | R |
| X | N | Y | I | E | L | D | I | L | B | Z |
| K | W | G | E | L | Q | F | K | I | L | X |
| J | W | O | R | K | P | L | A | N | M | R |
| S | E | Q | U | E | N | C | E | E | D | M |
| M | C | P | S | K | D | O | H | T | E | M |
| G | Q | U | A | N | T | I | T | Y | A | F |
| J | L | C | E | R | A | P | E | R | P | R |
| K | Y | U | M | G | C | C | G | N | U | T |
| S | T | M | M | D | C | T | S | C | T | P |
| K | I | G | K | S | X | M | B | L | O | Q |

YIELD · QUANTITY · METHOD · PREPARE · SEQUENCE · MEASURE · INGREDIENT · WORKPLAN

**Quick finish: Which word tells you how many servings the recipe should make? ______________________** *Revisit: 8.1 Recipe, yield and quantities*

## BUSY WORK 16 · Production timeline puzzle

Number the actions 1-8 to form a sensible high-level workplan. Exact timing is Teacher to confirm.

| Statement or action | Answer |
|---|---|
| Assemble and present the burger when components are ready. | |
| Read the final recipe, workplan and teacher directions. | |
| Complete the teacher-approved cleaning and pack-up routine. | |
| Prepare personal hygiene and the workstation. | |
| Record quality checks and any changes during production. | |
| Measure and organise ingredients and equipment. | |
| Complete preparation tasks in a safe, efficient order. | |
| Use the planned cooking or production method under supervision. | |

**Quick finish: Where would you include a checkpoint for a teacher decision? _________________________** *Revisit: 8.2 Workplans and production sequencing*

MODULE 8 · APPLY

# Project evidence · Plan production

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### FINAL RECIPE: YIELD, EXACT INGREDIENTS AND NUMBERED METHOD

#### FOOD ORDER: TOTAL QUANTITY, SUPPLY RESPONSIBILITY AND SPECIAL NOTE

#### WORKPLAN: SETUP, PREPARATION, PRODUCTION, ASSEMBLY AND CLEANING

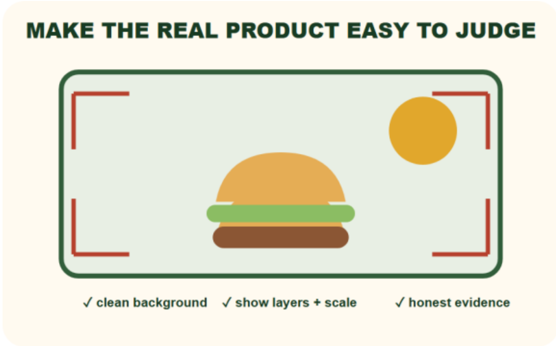
#### ONE QUALITY CHECKPOINT AND ONE TEACHER DECISION POINT

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 9 · READ

# 9. Produce and present

*Manage safety, timing and quality while making the final product.*




## 9.1 Follow, monitor, adapt

Use the workplan as a guide, not as an excuse to ignore evidence. Monitor appearance, texture, temperature controls and time. If a safe change is needed, pause, consult the teacher when required, record what changed and explain why.

## 9.2 Quality control

Quality control checks the product during production rather than waiting until the end. Examples include checking ingredient measurements, patty size, even preparation, browning, bun condition, layer order and workstation hygiene. Follow teacher instructions for checking that food is safely cooked.

## 9.3 Presentation and evidence

Presentation should support the intended user and make the burger's structure visible. Use a clean plate or board, remove distracting spills and avoid inedible decoration. Photograph the finished product in good light from an angle that shows its layers and scale. Record the final burger promptly. A photograph supports evidence but does not replace written evaluation.

**Key terms:** monitor · adapt · quality control · evidence · presentation · record

MODULE 9 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines monitored production?**

| | | |
|---|---|---|
| A following the plan while observing evidence and making safe, recorded adjustments | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A hide changes because they differ from the written plan | B wait until evaluation before considering it | C watch appearance, texture, time and required safety controls |

**3. Which statement best defines quality control?**

| | | |
|---|---|---|
| A checking important product and process features during production | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A wait until serving to notice every quality problem | B wait until evaluation before considering it | C check measurements, preparation, size, browning, layers and hygiene at planned points |

**5. Which statement best defines production evidence?**

| | | |
|---|---|---|
| A an honest record that makes the real product and process easy to judge | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A use an unrelated or heavily misleading image | B wait until evaluation before considering it | C present cleanly and photograph promptly in good light from a useful angle |

COMBINED GUIDED RESPONSE

1. Write a production log: what followed the plan, what changed, why it changed and how safety was maintained. 2. Record four quality-control checks for your final practical and the evidence you will collect for each. 3. Plan the photographs needed to prove production and final-product quality. Explain what each image must show.

> **Useful starters:** The production followed the plan when… I changed… because… Safety was maintained by… · At the… stage I will check… My evidence will be… · I will photograph… This image will show… and support…

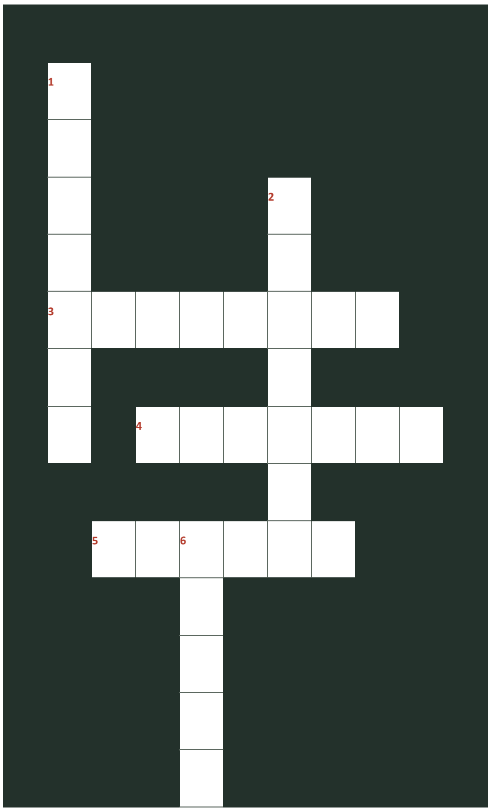
Self-check: ☐ all three ideas addressed ☐ subject vocabulary used ☐ claims linked to reasons or evidence

MODULE 9 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 17 · Quality-control crossword

Complete the crossword using words that describe effective monitoring and evidence.



Across: 3 Information used to support a judgement. | 4 A sensory quality felt in the mouth or by touch. | 5 Capture what happened so it can be reviewed later.
Down: 1 Practices that support safe food preparation. | 2 Observe progress during production. | 6 Compare what is happening with the plan or criterion.

**Quick finish: Which three answers form a useful cycle during production? ________________________** *Revisit: 9.1 Monitoring and adapting production*

## BUSY WORK 18 · Evidence-to-claim match

Match each evidence item A-F with the claim it can support 1-6.

| A-F | 1-6 |
|---|---|
| A. A labelled setup photograph | 1. The final appearance and assembly can be reviewed. |
| B. A timed workplan with changes noted | 2. Changes were made for stated reasons. |
| C. A final burger photograph | 3. The workstation and mise en place were prepared. |
| D. A teacher or peer feedback note | 4. Pack-up actions were completed. |
| E. A quality-check record | 5. An external response informed reflection. |
| F. A cleaning checklist | 6. Production was monitored against selected qualities. |

**Matches: A-__ B-__ C-__ D-__ E-__ F-__**

**Quick finish: Why is a photograph stronger when it has a label or caption? ________________________** *Revisit: 9.2 Capturing useful practical evidence*

MODULE 9 · APPLY

# Project evidence · Produce and present

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

### BEFORE: READINESS AND SETUP EVIDENCE

### DURING: QUALITY CHECKS, OBSERVATIONS AND SAFE CHANGES

### AFTER: FINAL PHOTOGRAPH AND WHAT IT PROVES

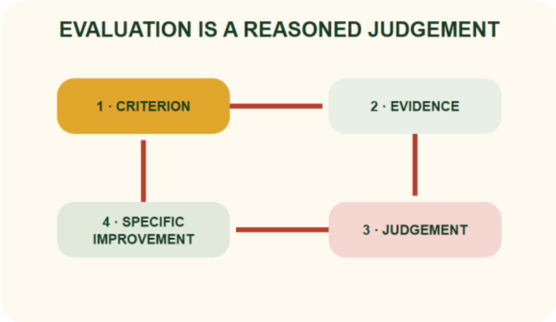
### FEEDBACK RECEIVED AND HOW I RESPONDED

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

MODULE 10 · READ

# 10. Evaluate and reflect

*Judge the solution with evidence and use the result to improve.*




## 10.1 Evaluate against criteria

Evaluation is a reasoned judgement, not a description of what you did. Return to each success criterion, state the result, cite evidence and explain the judgement. Evidence may include observations, measurements, photographs, user feedback and comparison with the design plan.

## 10.2 Consider the whole solution

Evaluate function, appearance, sensory properties, ergonomics, cost, resource use, safety, cultural suitability, ethical factors, sustainability and technical quality where relevant. Not every factor will be equally important, so explain priorities. Feedback is useful evidence, but one person's preference should not automatically outweigh the brief or safety requirements.

## 10.3 Reflect and improve

A useful improvement is specific and achievable: name what would change, how it would change and the expected effect. “Reduce the sauce from two tablespoons to one so the base stays firm” is stronger than “make it better”. Finish by reflecting on your planning, practical skills, independence and response to problems. Honest reflection shows learning.

> **Key terms:** criterion · evidence · judgement · sensory · function · reflection · improvement

MODULE 10 · CHECK AND EXPLAIN

# Check key ideas, then connect them

**1. Which statement best defines evidence-based evaluation?**

| | | |
|---|---|---|
| A a reasoned judgement of how well the solution met stated criteria | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**2. Which action best applies the theory?**

| | | |
|---|---|---|
| A repeat the production method instead of judging the result | B wait until evaluation before considering it | C state the criterion, cite evidence and explain the judgement |

**3. Which statement best defines whole-solution evaluation?**

| | | |
|---|---|---|
| A judging the relevant functional, sensory, safety, resource and social dimensions together | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**4. Which action best applies the theory?**

| | | |
|---|---|---|
| A allow one person's preference to override safety requirements | B wait until evaluation before considering it | C explain which factors mattered most for this brief and user |

**5. Which statement best defines specific improvement?**

| | | |
|---|---|---|
| A an achievable change that states what will change, how and with what expected effect | B a decorative feature with no effect on the product | C a task completed only after the final practical |

**6. Which action best applies the theory?**

| | | |
|---|---|---|
| A write only "make it better next time" | B wait until evaluation before considering it | C connect each weakness to a precise modification and predicted result |

### COMBINED GUIDED RESPONSE

1. Evaluate your final burger against at least six criteria. For each, give evidence and a clear judgement. 2. Explain which evaluation factors were most important for this project and how you balanced a trade-off. 3. Identify three specific improvements and reflect on your planning, practical skills, independence and response to problems.

**Useful starters:** For the criterion… my evidence is… Therefore, the burger… · The most important factors were… because… A trade-off occurred between… · Next time I will change… by… This should… I improved my ability to…

Self-check: ☐ all three ideas addressed ☐ subject vocabulary used ☐ claims linked to reasons or evidence

MODULE 10 · BUSY WORK

# Puzzle breaks: practise and revisit

## BUSY WORK 19 · Evaluation vocabulary find-a-word

Find the words used to turn description into a supported evaluation.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| D | Y | R | O | S | N | E | S | Y | C | P |
| R | T | C | E | L | F | E | R | R | A | S |
| U | T | K | C | M | E | F | I | H | Q | A |
| C | G | K | N | O | I | T | C | N | U | F |
| Y | T | N | E | M | E | G | D | U | J | E |
| A | V | K | D | R | V | M | M | I | J | T |
| H | U | B | I | T | O | L | A | L | J | Y |
| C | S | O | V | Q | R | M | W | A | N | Q |
| T | N | C | E | P | P | T | O | W | K | N |
| C | Y | W | A | P | M | A | F | J | I | O |
| Q | I | H | U | N | I | B | R | S | F | B |

CRITERION · EVIDENCE · JUDGEMENT · IMPROVE · REFLECT · FUNCTION · SAFETY · SENSORY

**Quick finish: Use three found words in one evaluation sentence: ________________________________** *Revisit: 10.1 Evaluating against criteria*

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## BUSY WORK 20 · Evaluation sentence builder

Write D (description), E (evidence), J (judgement) or I (improvement) beside each statement.

| Statement or action | Answer |
|---|---|
| The burger used a toasted bun, patty, salad and herb sauce. | |
| The tasting notes described a fresh herb aroma and balanced texture. | |
| The burger mostly met the stability criterion because the layers stayed aligned. | |
| Next time I would reduce the sauce quantity to limit moisture. | |
| The production record shows assembly finished later than planned. | |
| The final result was successful because it met three of the four tested criteria. | |

**Quick finish: Combine one evidence statement with one judgement using because: ___________________** *Revisit: 10.2 Reflection and specific improvements*

MODULE 10 · APPLY

# Project evidence · Evaluate and reflect

**Purpose:** Use the theory and puzzle practice to produce concise evidence for your Riv Burger design process.

#### CRITERION, EVIDENCE AND JUDGEMENT: FUNCTION AND ERGONOMICS

#### CRITERION, EVIDENCE AND JUDGEMENT: SENSORY AND APPEARANCE

#### CRITERION, EVIDENCE AND JUDGEMENT: SAFETY AND PRODUCTION

#### ONE SPECIFIC IMPROVEMENT AND EXPECTED EFFECT

**Evidence boundary:** Ask your teacher how this paper evidence should be stored or submitted. Practical arrangements and formal submission remain Teacher to confirm.

ANSWER KEY · KNOWLEDGE

# Knowledge-check answers · 1 of 4

| Modul e | Theory section | Q | Answer and revisit |
|---|---|---|---|
| **1** | 1.1 Design situation and brief | 1 | A · a clear design task developed from a need or opportunity · Revisit 1.1 Design situation and brief |
| **1** | 1.1 Design situation and brief | 2 | C · identify the product, intended user and important qualities before developing ideas · Revisit 1.1 Design situation and brief |
| **1** | 1.2 Users and needs | 3 | A · something the customer or producer requires from the solution · Revisit 1.2 Users and needs |
| **1** | 1.2 Users and needs | 4 | C · balance flavour, appearance, value, convenience, size and production needs · Revisit 1.2 Users and needs |
| **1** | 1.3 Criteria and constraints | 5 | A · a specific quality that can be checked to judge design success · Revisit 1.3 Criteria and constraints |
| **1** | 1.3 Criteria and constraints | 6 | C · write measurable or observable criteria and list realistic constraints · Revisit 1.3 Criteria and constraints |
| **2** | 2.1 Personal hygiene | 1 | A · actions that prevent a food handler transferring contamination to food · Revisit 2.1 Personal hygiene |
| **2** | 2.1 Personal hygiene | 2 | C · secure hair, cover cuts, wear required protection and wash and dry hands correctly · Revisit 2.1 Personal hygiene |
| **2** | 2.2 Cross-contamination | 3 | A · the transfer of harmful microorganisms or allergens between food, people, tools or surfaces · Revisit 2.2 Cross-contamination |
| **2** | 2.2 Cross-contamination | 4 | C · separate raw and ready-to-eat foods and use cleaned, sanitised equipment · Revisit 2.2 Cross-contamination |
| **2** | 2.3 Hazards and controls | 5 | A · an action that reduces the chance or severity of harm · Revisit 2.3 Hazards and controls |
| **2** | 2.3 Hazards and controls | 6 | C · match each identified hazard with a specific control · Revisit 2.3 Hazards and controls |
| **3** | 3.1 Mise en place | 1 | A · putting ingredients, equipment and the work area in place before production · Revisit 3.1 Mise en place |
| **3** | 3.1 Mise en place | 2 | C · read the method, prepare the bench, collect tools and measure ingredients · Revisit 3.1 Mise en place |
| **3** | 3.2 Selecting tools | 3 | A · choosing equipment because its function suits the required task · Revisit 3.2 Selecting tools |

ANSWER KEY · KNOWLEDGE

# Knowledge-check answers · 2 of 4

| Module | Theory section | Q | Answer and revisit |
|---|---|---|---|
| **3** | 3.2 Selecting tools | 4 | C · match the cook’s knife, small knife, scale, spoons or spatula to its intended job · Revisit 3.2 Selecting tools |
| **3** | 3.3 A clean workflow | 5 | A · an ordered movement from preparation through cooking, assembly and cleaning · Revisit 3.3 A clean workflow |
| **3** | 3.3 A clean workflow | 6 | C · clean as you go, manage spills and return ingredients to safe storage · Revisit 3.3 A clean workflow |
| **4** | 4.1 A layered food system | 1 | A · interacting burger parts in which each layer affects the whole product · Revisit 4.1 A layered food system |
| **4** | 4.1 A layered food system | 2 | C · give every layer a clear function and arrange it deliberately · Revisit 4.1 A layered food system |
| **4** | 4.2 Sensory properties | 3 | A · a product feature detected through appearance, aroma, flavour, texture or sound · Revisit 4.2 Sensory properties |
| **4** | 4.2 Sensory properties | 4 | C · use exact sensory vocabulary and plan complementary contrasts · Revisit 4.2 Sensory properties |
| **4** | 4.3 Function and stability | 5 | A · the ability of the assembled product to remain together and comfortable to hold · Revisit 4.3 Function and stability |
| **4** | 4.3 Function and stability | 6 | C · drain vegetables, match component sizes and position wet or slippery layers carefully · Revisit 4.3 Function and stability |
| **5** | 5.1 From primary production to plate | 1 | A · the connected stages from primary production through processing, transport, retail and service · Revisit 5.1 From primary production to plate |
| **5** | 5.1 From primary production to plate | 2 | C · link each ingredient to its agricultural industry and later journey stages · Revisit 5.1 From primary production to plate |
| **5** | 5.2 Local and seasonal choices | 3 | A · a claim supported by a defined region and evidence of producer or ingredient origin · Revisit 5.2 Local and seasonal choices |
| **5** | 5.2 Local and seasonal choices | 4 | C · define the region and verify producer, origin, distance and season · Revisit 5.2 Local and seasonal choices |
| **5** | 5.3 Ethical and environmental factors | 5 | A · a choice that considers environmental, ethical and practical effects across the production system · Revisit 5.3 Ethical and environmental factors |
| **5** | 5.3 Ethical and environmental factors | 6 | C · compare evidence about soil, water, welfare, energy, packaging, transport and waste · Revisit 5.3 Ethical and environmental factors |

ANSWER KEY · KNOWLEDGE

# Knowledge-check answers · 3 of 4

| Module | Theory section | Q | Answer and revisit |
|---|---|---|---|
| **6** | 6.1 Divergent thinking | 1 | A · generating several meaningfully different responses before selecting one · Revisit 6.1 Divergent thinking |
| **6** | 6.1 Divergent thinking | 2 | C · change filling, bun, herbs, vegetables, sauce, form or flavour direction · Revisit 6.1 Divergent thinking |
| **6** | 6.2 Annotating concepts | 3 | A · a note that explains the reason, function or expected effect of a design decision · Revisit 6.2 Annotating concepts |
| **6** | 6.2 Annotating concepts | 4 | C · label ingredients, layer order, sensory effect, source, safety and user benefit · Revisit 6.2 Annotating concepts |
| **6** | 6.3 Choosing with evidence | 5 | A · choosing a concept by comparing it with criteria and explaining trade-offs · Revisit 6.3 Choosing with evidence |
| **6** | 6.3 Choosing with evidence | 6 | C · score options, give reasons and identify a needed refinement · Revisit 6.3 Choosing with evidence |
| **7** | 7.1 What plants need | 1 | A · a condition or resource required for healthy growth · Revisit 7.1 What plants need |
| **7** | 7.1 What plants need | 2 | C · manage light, water, air, temperature, nutrients and root conditions together · Revisit 7.1 What plants need |
| **7** | 7.2 Growing and harvesting herbs | 3 | A · collecting correctly identified healthy plant material while protecting safety and future growth · Revisit 7.2 Growing and harvesting herbs |
| **7** | 7.2 Growing and harvesting herbs | 4 | C · gain permission, identify the herb, use clean tools and wash produce as directed · Revisit 7.2 Growing and harvesting herbs |
| **7** | 7.3 Sustainability in practice | 5 | A · meeting present needs while protecting future environmental, social and economic wellbeing · Revisit 7.3 Sustainability in practice |
| **7** | 7.3 Sustainability in practice | 6 | C · water efficiently, care for soil, plan harvests and measure ingredients · Revisit 7.3 Sustainability in practice |
| **8** | 8.1 A production-ready recipe | 1 | A · exact ingredients, quantities, yield and an ordered method that another cook can follow · Revisit 8.1 A production-ready recipe |
| **8** | 8.1 A production-ready recipe | 2 | C · use consistent units, preparation words and safe sequence · Revisit 8.1 A production-ready recipe |
| **8** | 8.2 Food order and resources | 3 | A · a totalled list of ingredients required for the planned yield · Revisit 8.2 Food order and resources |

ANSWER KEY · KNOWLEDGE

# Knowledge-check answers · 4 of 4

| Module | Theory section | Q | Answer and revisit |
|---|---|---|---|
| **8** | 8.2 Food order and resources | 4 | C · check supplies, total quantities and flag allergens or storage needs · Revisit 8.2 Food order and resources |
| **8** | 8.3 Workplan | 5 | A · a timed map of activities, equipment and ingredients across the practical · Revisit 8.3 Workplan |
| **8** | 8.3 Workplan | 6 | C · sequence tasks, identify safe overlaps and protect evidence and cleaning time · Revisit 8.3 Workplan |
| **9** | 9.1 Follow, monitor, adapt | 1 | A · following the plan while observing evidence and making safe, recorded adjustments · Revisit 9.1 Follow, monitor, adapt |
| **9** | 9.1 Follow, monitor, adapt | 2 | C · watch appearance, texture, time and required safety controls · Revisit 9.1 Follow, monitor, adapt |
| **9** | 9.2 Quality control | 3 | A · checking important product and process features during production · Revisit 9.2 Quality control |
| **9** | 9.2 Quality control | 4 | C · check measurements, preparation, size, browning, layers and hygiene at planned points · Revisit 9.2 Quality control |
| **9** | 9.3 Presentation and evidence | 5 | A · an honest record that makes the real product and process easy to judge · Revisit 9.3 Presentation and evidence |
| **9** | 9.3 Presentation and evidence | 6 | C · present cleanly and photograph promptly in good light from a useful angle · Revisit 9.3 Presentation and evidence |
| **10** | 10.1 Evaluate against criteria | 1 | A · a reasoned judgement of how well the solution met stated criteria · Revisit 10.1 Evaluate against criteria |
| **10** | 10.1 Evaluate against criteria | 2 | C · state the criterion, cite evidence and explain the judgement · Revisit 10.1 Evaluate against criteria |
| **10** | 10.2 Consider the whole solution | 3 | A · judging the relevant functional, sensory, safety, resource and social dimensions together · Revisit 10.2 Consider the whole solution |
| **10** | 10.2 Consider the whole solution | 4 | C · explain which factors mattered most for this brief and user · Revisit 10.2 Consider the whole solution |
| **10** | 10.3 Reflect and improve | 5 | A · an achievable change that states what will change, how and with what expected effect · Revisit 10.3 Reflect and improve |
| **10** | 10.3 Reflect and improve | 6 | C · connect each weakness to a precise modification and predicted result · Revisit 10.3 Reflect and improve |

ANSWER KEY · BUSY WORK

# Busy Work answers · 1 of 2

Check only after attempting each activity. For open decisions, compare the quality of the reason as well as the short answer.

| # | Activity | Answer | Revisit |
|---|---|---|---|
| 1 | Design language find-a-word | CONSTRAINT R10C5-R1C5; CRITERIA R6C2-R6C9; FUNCTION R1C1-R1C8; EVIDENCE R3C3-R3C10; DESIGN R9C7-R4C2; BRIEF R6C10-R2C10; USER R8C8-R5C5; NEED R1C8-R4C11 | 1.1 Design situation and brief |
| 2 | Brief sorter | 1=U; 2=CO; 3=CR; 4=N; 5=CR; 6=CO | 1.2 Criteria and constraints |
| 3 | Food-safety crossword | 1A CLEAN; 1D CONTROL; 2D SEPARATE; 3D HYGIENE; 4D HAZARD; 5A ALLERGEN | 2.1 Hygiene, allergens and contamination |
| 4 | Hazard-control match | A-3, B-5, C-6, D-4, E-1, F-2 | 2.2 Managing hazards and risk |
| 5 | Mise en place shuffle | 1=4; 2=1; 3=7; 4=2; 5=3; 6=6; 7=5; 8=8 | 3.1 Mise en place and workflow |
| 6 | Tool-to-job match | A-4, B-1, C-5, D-2, E-3, F-6 | 3.2 Tools, equipment and safe technique |
| 7 | Burger layer find-a-word | STABILITY R10C8-R2C8; FILLING R2C11-R8C5; TEXTURE R9C5-R9C11; FLAVOUR R2C11-R8C11; SAUCE R10C8-R10C4; FRESH R10C10-R6C10; AROMA R8C8-R4C4; BUN R8C9-R10C9 | 4.1 Burger systems and layer functions |
| 8 | Stack stability trouble-shooter | 1=Review layer order, moisture and grip between layers.; 2=Review wet ingredients, sauce quantity and assembly timing.; 3=Add a suitable contrasting texture while keeping the brief in mind.; 4=Reduce or reorganise layers to improve ergonomics. | 4.2 Sensory and functional analysis |
| 9 | Ingredient journey crossword | 1D SEASONAL; 2D EVIDENCE; 3A PROCESS; 4A PRODUCER; 5A RETAIL; 6D LOCAL | 5.1 From farm to plate |
| 10 | Claim detective | 1=S; 2=V; 3=S; 4=V; 5=S; 6=V | 5.2 Local, seasonal and sustainable choices |

ANSWER KEY · BUSY WORK

# Busy Work answers · 2 of 2

Check only after attempting each activity. For open decisions, compare the quality of the reason as well as the short answer.

| # | Activity | Answer | Revisit |
|---|---|---|---|
| 11 | Concept variation find-a-word | VEGETABLE R2C11-R2C3; STRUCTURE R1C7-R9C7; FILLING R8C9-R2C9; FLAVOUR R2C1-R8C7; CONCEPT R5C5-R11C5; SAUCE R5C6-R9C6; HERB R1C9-R4C6; BUN R9C3-R7C5 | 6.1 Generating design options |
| 12 | Concept quick-test | 1=S; 2=W; 3=E; 4=S; 5=W | 6.2 Selecting and justifying a concept |
| 13 | Plant-needs crossword | 1D HARVEST; 2D AIR; 3D WATER; 4A NUTRIENTS; 5D SOIL; 6A LIGHT; 7A ROOTS; 8D SPACE | 7.1 Plant needs and herb growth |
| 14 | Garden problem solver | A-2, B-5, C-3, D-1, E-6, F-4 | 7.2 Garden planning and responsible harvesting |
| 15 | Recipe language find-a-word | INGREDIENT R1C1-R10C10; QUANTITY R7C2-R7C9; SEQUENCE R5C1-R5C8; WORKPLAN R4C2-R4C9; PREPARE R8C10-R8C4; MEASURE R9C4-R3C4; METHOD R6C11-R6C6; YIELD R2C3-R2C7 | 8.1 Recipe, yield and quantities |
| 16 | Production timeline puzzle | 1=7; 2=1; 3=8; 4=2; 5=6; 6=3; 7=4; 8=5 | 8.2 Workplans and production sequencing |
| 17 | Quality-control crossword | 1D HYGIENE; 2D MONITOR; 3A EVIDENCE; 4A TEXTURE; 5A RECORD; 6D CHECK | 9.1 Monitoring and adapting production |
| 18 | Evidence-to-claim match | A-3, B-2, C-1, D-5, E-6, F-4 | 9.2 Capturing useful practical evidence |
| 19 | Evaluation vocabulary find-a-word | CRITERION R1C10-R9C2; JUDGEMENT R5C10-R5C2; EVIDENCE R9C4-R2C4; FUNCTION R4C11-R4C4; IMPROVE R11C6-R5C6; REFLECT R2C8-R2C2; SENSORY R1C8-R1C2; SAFETY R2C11-R7C11 | 10.1 Evaluating against criteria |
| 20 | Evaluation sentence builder | 1=D; 2=E; 3=J; 4=I; 5=E; 6=J | 10.2 Reflection and specific improvements |

EVIDENCE GUIDE

# Assessment and evidence overview

The authorised source task asks students to design and produce a signature burger using locally grown produce and home-grown herbs, communicate the design process and evaluate the final solution.

| Component | Source marks | Evidence represented in this workbook |
|---|---|---|
| Folio | 25 | Research, design criteria and concepts, recipe, planning and evaluation. |
| Practical skills | 60 | Safe and hygienic work, preparation, equipment use, production management and cleaning. |
| Final design | 15 | Function, aesthetics, creativity, ergonomics, ethical and environmental considerations and time management. |
| **Total** | **100** | **Combined design, production and evaluation evidence.** |

TEN-MODULE EVIDENCE CHAIN

| | |
|---|---|
| **1 Challenge and brief** | **2 Safe food practices** |
| **3 Tools and workflow** | **4 Burger anatomy** |
| **5 Ingredients and agriculture** | **6 Develop your design** |
| **7 Garden to kitchen** | **8 Plan production** |
| **9 Produce and present** | **10 Evaluate and reflect** |

**Teacher to confirm:** Whether the source mark allocation applies to the current class, plus current due dates, lesson timing, supplied ingredients and submission arrangements.

MY STRONGEST EVIDENCE AND WHERE IT APPEARS

FINAL REVIEW

## Final compact-workbook check

Use this page to confirm that your shortened workbook still tells a connected design story from the original need to the evaluated solution.

- ☐ I completed the learning and evidence pages for all ten modules.
- ☐ I attempted the knowledge checks before using the answer key.
- ☐ I completed both Busy Work activities in each module.
- ☐ My brief, intended user, criteria and constraints are clear.
- ☐ My safety decisions identify hazards and specific controls.
- ☐ My concepts are genuinely different and my final choice uses evidence.
- ☐ My recipe, food order and workplan agree with one another.
- ☐ My practical record shows monitoring, safe changes, feedback and photographs.
- ☐ My evaluation uses criteria, evidence, judgements and specific improvements.
- ☐ I followed my teacher's current safety, allergen and submission instructions.

THE STRONGEST CONNECTION ACROSS MY DESIGN PROCESS IS

THE NEXT THING I NEED TO IMPROVE OR CONFIRM IS

**Keep the boundary clear:** This compact workbook supports learning and evidence. Your teacher confirms what must be formally submitted and how.